

Hansheng Liu

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EDUCATION

University of Illinois Urbana-Champaign

- Master of **Computer Science** Aug 2024 – Dec 2025 at Urbana, IL
 - GPA: 3.74 / 4.00 | Teaching Assistantship - Full Tuition Scholarship
- Bachelor of Science in **Computer Science & Mathematics** Aug 2020 – May 2024 at Urbana, IL
 - GPA: 3.96 / 4.00 | Highest Distinction
 - Additional Major: Economics. Minor: Statistics, Computational Science & Engineering

SKILLS

Programming Languages: Java, C#, C++, Python, SQL, JavaScript, TypeScript

Frameworks: Spring Boot, Spring Cloud, Spring Security, Hibernate, Angular, React, JUnit, Mockito

Middleware & Database: MySQL, SQL Server, Oracle, PostgreSQL, MongoDB, Redis, Kafka, RabbitMQ

DevOps & Cloud: AWS (Lambda, EC2, RDS, S3), Docker, Kubernetes, Jenkins, NGINX, Git, Linux

WORK EXPERIENCE

Orchestrade

Software Development Engineer

Apr 2026 – Present at San Mateo, CA

- Develop end-to-end features for a multi-asset trading and risk management platform in **C# / .NET**, spanning desktop UI, application services, market-data processing, persistence, and relational databases.
- Implement pricing and settlement logic for complex physical and financial energy products, translating trading requirements such as production thresholds, year-to-date settlement, and positive-volume conditions into production domain logic.
- Build configurable numerical-precision support for trade pricing fields, integrating system configuration with reusable grid, layout, and editor components while preserving backward-compatible defaults.
- Extend product and market-data workflows across a mature enterprise codebase, debugging date/time-zone behavior, historical data selection, and cross-layer interactions between client and server components.
- Maintain cross-database functionality across **SQL Server**, **PostgreSQL**, and **Oracle**, implementing schema changes and migration logic while ensuring consistent application behavior across environments.

Amazon

Software Development Engineer Intern

May – Aug 2025 at Bellevue, WA

- Engineered a **Java**-based natural-language interface that converted user queries into structured API calls and retrieved results from internal services, reducing manual effort by 90%.
- Refactored local MCP servers into **AWS Lambda** functions, enabling authentication, automatic scaling, and easy access.
- Built **JUnit** and **Mockito** test suites for decoupled components, achieving 100% code coverage.
- Authored technical design documents, evaluating architectural tradeoffs and securing cross-functional stakeholder buy-in.

Leyan Technologies

R&D Engineer

Jun – Aug 2024 at Shanghai, China

- Developed an AI framework validation system, automating correctness verification and performance benchmarking.
- Spearheaded the implementation of the web backend (**Spring Boot**), test-runner (**Python**), and database (**MySQL**).

Zhejiang Century Huatong Group

Software Development Engineer

Jun – Aug 2023 at Shanghai, China

- Engineered and deployed 10+ core modules for the Office Automation system using **C#**, ensuring 100% on-time delivery.
- Integrated taxi API into corporate app, streamlining employee reimbursement workflows and enhancing expense oversight.

Illinois Mathematics Lab

Researcher

Jan – Dec 2023 at Urbana, IL

- Engineered a **C++** simulation system, utilizing multithreading and optimized algorithms to reduce runtime by over 80%.
- Mined frequent patterns and applied multiple deep-learning algorithms to derive insights from the data.

PROJECT HIGHLIGHTS

HR Onboarding Portal

Spring Boot, Spring Cloud, Spring Security, Kafka, JWT, AWS (RDS, S3), MySQL, MongoDB

- Architected a microservices-based HR onboarding portal with a **Spring Boot** backend and **React** frontend.
- Engineered a centralized cloud infrastructure utilizing **Netflix Eureka** for service discovery, **Spring Cloud API Gateway** for request routing, and **Spring Cloud Config** for distributed configuration management.
- Implemented **Apache Kafka** for asynchronous inter-service communication, scaling to handle high-volume requests.
- Secured the microservices cluster by developing a standalone Authentication Server with **Spring Security** and **JWT**.
- Integrated **AWS RDS (MySQL)** and **MongoDB Atlas** for data persistence, and **AWS S3** for secure document storage.
- Automated testing and deployment to **AWS EC2** via **GitHub Actions** CI/CD pipelines.

Online Shopping Application

Spring Boot, Hibernate, MySQL, Spring Security, Angular

- Architected a full-stack RESTful e-commerce platform with a **Spring Boot** backend and **Angular** frontend.
- Implemented a strict layered architecture and **Hibernate (Criteria API)** for dynamic database queries.
- Secured the application using **Spring Security** and **JWT** for stateless authentication and role-based authorization.
- Maintained ACID compliance during inventory changes by implementing robust transaction management and **Spring AOP**.

Chat Master

NGINX, React, Python

- Built a distributed chat service behind **NGINX**, supporting 8,000+ concurrent connections in load testing.

RESEARCH

Packing Low-diameter Spanning Trees

First author. Advised by Rhea Jain

2025

- Graph Algorithm. Randomized Algorithm. Approximation Algorithm.

Hop-constrained Shortest Path Queries in Planar Graphs

First author. Advised by Prof. Jeff Erickson

2025

- Data Structure. Graph Algorithm.

Length-constrained Minimum Spanning Tree

First author. Advised by Rhea Jain & Prof. Chandra Chekuri

2025

- Graph Algorithm. Approximation Algorithm. Derandomization.

Chain of Summary with persistently decaying entropy in RAG

First author. Advised by Prof. Jiawei Han

2024

- Text Mining. Post-RAG. LLM.

Scale-Free First-Passage Percolation

Student author. Mentored by Prof. Partha S. Dey

2023

- Stochastic Processes. Approximation Algorithm. Deep Learning.
- 2024 IML Research Award.

Rumor Propagation on a Spread-out Line Graph

Student author. Mentored by Prof. Partha S. Dey

2023

- Stochastic Processes.
- 2023 IGL Research Award.